

Dual-Port Live 3D Rendering Architecture

Phase 16 Theoretical Foundation Document

VSEPR-SIM — Atomistic View (Port 9999) and Analysis View (Port 10001)

VSEPR-SIM Development

Version 1.0.0 — Phase 16

Contents

1 Scope and Purpose

This document establishes the theoretical and architectural foundation for the dual-port live visualisation stream server introduced in Phase 16. The system answers two operationally distinct questions simultaneously:

- **Port 9999 — Atomic View:** “*What is the solver doing right now?*” Fast, low-latency, tactical live viewer.
- **Port 10001 — Analysis View:** “*What has the solver found, tested, compared, and inferred?*” Heavy interpretive layer: history, candidates, property panels, deeper 3D scenes.

Both views share one simulation engine and communicate via a newline-delimited JSON (NDJSON) wire protocol over raw TCP sockets.

2 Atom Record Definition

The fundamental unit of the Atomic View frame is the per-atom record:

$$\mathcal{A}_i(t) = (\text{id}_i, Z_i, \mathbf{r}_i(t), \mathbf{v}_i(t), q_i, s_i, \chi_i) \quad (1)$$

where:

- id_i — unique integer atom index
- Z_i — species or atomic number
- $\mathbf{r}_i(t) = (x_i, y_i, z_i)$ — position in Å
- $\mathbf{v}_i(t) = (v_x, v_y, v_z)$ — velocity in m s^{-1} (Maxwell-Boltzmann sample or pseudo-motion vector)
- q_i — charge or effective scalar property
- $s_i \in \{0, 1, 2, 3\}$ — state tag (0 = normal, 1 = active/focus, 2 = defect, 3 = selected)
- $\chi_i \in [0, 1]$ — local fit or confidence score, defined below in (??)

The complete live atomistic state at time t is the set:

$$\mathcal{A}(t) = \{\mathcal{A}_1(t), \mathcal{A}_2(t), \dots, \mathcal{A}_N(t)\} \quad (2)$$

2.1 Confidence / Instability Score

The local confidence score χ_i is computed from the ratio of the atom’s instantaneous kinetic energy to a reference thermal scale:

$$\chi_i = \min\left(1, \frac{K_i}{3 K_{\text{avg}}(T)}\right), \quad K_{\text{avg}}(T) = \frac{3}{2} k_B T \quad (3)$$

where $K_i = \frac{1}{2} m |\mathbf{v}_i|^2$ (per molecule) and $k_B = 1.380649 \times 10^{-23} \text{ J K}^{-1}$.

A value near 0 indicates a thermally cold, well-placed atom; near 1 indicates a hot, potentially defective or unstable region.

3 Focus Subset and Active Work Region

The solver does not treat all atoms equivalently at every iteration. A focus subset is defined:

$$\mathcal{F}(t) \subseteq \mathcal{A}(t) \quad (4)$$

At minimum, $\mathcal{F}(t)$ contains the currently active atom $\mathcal{A}_{f(t)}$ where $f(t)$ is the focus index. In the gas2 implementation $f(t)$ is taken as $f(t) = t \bmod N$, cycling through all atoms once per full sweep.

The viewer renders $\mathcal{F}(t)$ with a distinct highlight: a glow ring of radius $r_{\text{px}} + 4$ pixels around the focus atom, coloured with the accent teal (hex #03dac6).

4 Lattice and Cell Overlay

When a crystalline interpretation is active, the current working lattice is:

$$\mathcal{L}(t) = \{ \mathbf{a}(t), \mathbf{b}(t), \mathbf{c}(t) \} \quad (5)$$

This is rendered as a wireframe box with 12 dashed edges, one edge per pair of opposite-face corners:

$$\text{vertices} = \{ \mathbf{0}, \mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{a} + \mathbf{b}, \mathbf{a} + \mathbf{c}, \mathbf{b} + \mathbf{c}, \mathbf{a} + \mathbf{b} + \mathbf{c} \} \quad (6)$$

In the gas2 phase, $\mathcal{L}$ is inactive (`active = false`) and the wireframe is suppressed. Lattice parameters a, b, c are estimated from the ideal-gas molar volume:

$$a = b = c = \left(\frac{V_m}{N_A} \right)^{1/3} \times 10^{10} \text{ \AA} \quad (7)$$

5 Bond and Interaction Layer

Pairwise interaction records are defined as:

$$\mathcal{B}_{ij}(t) = (i, j, w_{ij}(t), \tau_{ij}) \quad (8)$$

where w_{ij} is the interaction weight and τ_{ij} the type index:

τ_{ij}	Mode	Colour
0	Chemical bond	#88aacc (steel blue)
1	Neighbor graph	#444455 (dim)
2	Crystal adjacency	#aa66cc (violet)
3	Force vector	#cc8844 (amber)

In the gas2 atomic view, mode 1 (neighbor graph) is used with cutoff:

$$r_{\text{cut}} = 1.5 d_{\text{spacing}} \quad d_{\text{spacing}} = 3.5 d_{\text{kin}} \quad (9)$$

and weight:

$$w_{ij} = 1 - \frac{r_{ij}}{r_{\text{cut}}}, \quad r_{ij} = |\mathbf{r}_j - \mathbf{r}_i| \quad (10)$$

6 Live HUD State Vector

The onscreen HUD displays a scalar state vector:

$$\mathcal{R}(t) = (E(t), T(t), \phi(t), N_d(t), \epsilon(t)) \quad (11)$$

where:

- $E(t)$ — total kinetic energy in Hartree
- $T(t)$ — current temperature in K
- $\phi(t)$ — phase guess (gas, vapor, liquid, supercritical, solid)
- $N_d(t)$ — defect count
- $\epsilon(t)$ — convergence residual:

$$\epsilon(t) = \left| \frac{\langle K \rangle / N}{\frac{3}{2} k_B T} - 1 \right| \quad (12)$$

A value of $\epsilon = 0$ indicates perfect equipartition; values above ~ 0.1 signal under-thermalised or anomalous regions.

6.1 Phase Identification from Critical Temperature

Phase $\phi(t)$ is inferred from the reduced temperature $T^* = T/T_c$:

$$\phi(t) = \begin{cases} \text{gas} & T^* > 1.20 \\ \text{supercritical} & 0.90 < T^* \leq 1.20 \\ \text{vapor} & 0.50 < T^* \leq 0.90 \\ \text{liquid} & T^* \leq 0.50 \end{cases} \quad (13)$$

7 Colour Modes

The Atomic View supports five colour modes:

Index	Mode	Formula
0	Element (CPK)	RGB from Z-indexed table
1	Energy	Interpolate (#2244aa→#ff3300) by K_i
2	Velocity	HSV hue: $h = 0.65 - 0.65 v_i/1000$
3	Defect	#cf6679 if $s_i = 2$, else accent
4	χ	Interpolate (#224488→#ffaa00) by χ_i

Energy tinting also modulates the CPK colour:

$$r += 0.4 \chi_i, \quad b -= 0.4 \chi_i \quad (14)$$

giving a hot (red-shifted) appearance for high- χ atoms.

8 3D Projection Engine

A lightweight Euler-angle perspective projection is used in both Python viewers.

8.1 Rotation

Two successive Euler rotations are applied:

$$\text{X-rotation: } \mathbf{v}' = R_x(\theta_x) \mathbf{v}, \quad R_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix} \quad (15)$$

$$\text{Y-rotation: } \mathbf{v}'' = R_y(\theta_y) \mathbf{v}', \quad R_y(\theta) = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix} \quad (16)$$

8.2 Perspective projection

Screen coordinates are computed by:

$$(s_x, s_y) = \left(c_x + \frac{v_x'' f}{v_z'' + f}, c_y - \frac{v_y'' f}{v_z'' + f} \right), \quad f = 600 \text{ px (focal length)} \quad (17)$$

where (c_x, c_y) is the canvas centre and depth $z = v_z'' + f$.

8.3 Depth sorting

Atoms are sorted back-to-front by z before rendering (painter's algorithm), ensuring correct visual occlusion without a depth buffer:

$$\text{draw order} \propto z \text{ (ascending)} \quad (18)$$

Atom screen radius scales with distance:

$$r_{\text{px}} = \text{clamp} \left(\frac{8 s_{\text{atom}} \cdot f}{z}, 3, 30 \right) \quad (19)$$

where s_{atom} is the user-controlled atom scale factor.

9 Analysis View: Historical Metrics

The Analysis View maintains a rolling history buffer of length $H = 300$ for the following time-series quantities:

$$\{E(t_k), T(t_k), \rho(t_k), \epsilon(t_k), \Phi(t_k)\}_{k=1}^H \quad (20)$$

where:

- $E(t_k)$ — translational KE per molecule in Hartree
- $T(t_k)$ — current temperature K
- $\rho(t_k) = M_{\text{mol}} N_A / V_m [\text{g}/\text{cm}^3]$ — density from ideal EOS
- $\epsilon(t_k)$ — convergence residual (equipartition)
- $\Phi(t_k)$ — phase score (placeholder for future objective function)

History is accumulated in a `std::deque<double>` capped at H ; analysis frames are published every fifth atomic cycle, keeping the analysis stream at $\sim \frac{1}{5}$ the atomic frame rate.

10 Speed Distribution: Maxwell–Boltzmann Bins

The Analysis View renders the analytical Maxwell–Boltzmann speed distribution in $N_b = 64$ equal-width bins over $v \in [0, 2.5 v_{\text{rms}}]$:

$$f(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 \exp\left(-\frac{m v^2}{2 k_B T}\right) \quad (21)$$

The three characteristic speeds are marked on the plot:

$$v_{\text{mp}} = \sqrt{\frac{2k_B T}{m}} \quad (22)$$

$$\bar{v} = \sqrt{\frac{8k_B T}{\pi m}} \quad (23)$$

$$v_{\text{rms}} = \sqrt{\frac{3k_B T}{m}} \quad (24)$$

The co-plotted KE distribution is:

$$g_K(v) = f(v) \frac{\frac{1}{2} m v^2}{E_h} \quad (\text{in Hartree units}) \quad (25)$$

rendered as an inferno-palette histogram.

11 Candidate Comparison Panel

The solver tests M candidate structural interpretations:

$$\mathcal{C} = \{C_1, C_2, \dots, C_M\} \quad (26)$$

Each candidate carries:

$$C_j = (E_j, \rho_j, Z_j, \Pi_j) \quad (27)$$

where E_j is the candidate energy (Hartree), ρ_j its density (g/cm³), Z_j the compressibility factor from the corresponding EOS, and Π_j the macro-property vector.

In the gas2 implementation, the three EOS variants serve as the initial candidate set:

j	Label	EOS
1	Ideal	$PV = nRT$
2	VdW	$(P + a/V_m^2)(V_m - b) = RT$
3	Redlich-K	$P = RT/(V_m - b) - a/(T^{1/2}V_m(V_m + b))$

The compressibility deviation $Z_j - 1$ expresses non-ideality; the candidate with the smallest $|Z_j - 1|$ is the thermodynamically consistent choice.

12 Macro-Property Panel

The property panel exposes the state vector:

$$P(t) = [\rho, K_{\text{bulk}}, G, E_Y, \nu, \alpha_T, \theta_D, \sigma_y, k_{\text{eff}}, c_s]^T \quad (28)$$

In the gas2 phase, solid-elastic properties (G, E_Y, ν, σ_y) are zero; the estimated quantities are:

- $\rho = M_{\text{mol}} N_A / V_m$ (g/cm³)
- $K_{\text{bulk}} = P$ (GPa, directly from EOS)
- $\alpha_T = 1/T$ (ideal gas thermal expansion, 10⁻⁶ K⁻¹)
- $c_s = \sqrt{\gamma R T / M}$ (adiabatic sound speed)
- $T_{\text{melt}} \approx 0.7 T_c$ (crude liquid–solid estimate)
- $k_{\text{eff}} = k_{\text{kin}}$ (species thermal conductivity at 300 K)

Property bars are coloured with the Inferno palette at normalised value:

$$t_j = \min\left(1, \frac{|P_j|}{P_j^{\text{max}}}\right) \quad (29)$$

where P_j^{max} is a display-range ceiling per property.

13 Wire Protocol and Frame Rate

Frames are transmitted as newline-delimited JSON (NDJSON) over raw TCP. Each frame is a single complete JSON object followed by a newline character `\n`; clients read line-by-line.

13.1 Frame publishing rates

Port	Frame type	Target rate
9999	<code>AtomicFrame</code>	15 fps (default)
10001	<code>AnalysisFrame</code>	2 fps (default)

The downsampled publish rule for analysis frames:

$$t_k \mapsto t_{k+m}, \quad m = 5 \quad (30)$$

reduces network and render overhead while retaining full historical detail in the buffer.

13.2 Server thread model

The `VizServer` class spawns two listener threads: one for each port. Each accepted client runs in a detached thread. The latest-frame pattern (mutex-guarded push + poll) decouples the simulation loop from client delivery cadence:

- Simulation thread: calls `push_atomic()` at full solver rate
- Client thread: polls latest frame and sends at target fps interval
- No frame queue: clients always receive the *current* state, never stale frames older than 1/fps

13.3 Socket setup

```
1 // Pseudocode: server init (POSIX path shown)
2 int fd = socket(AF_INET, SOCK_STREAM, 0);
3 setsockopt(fd, SOL_SOCKET, SO_REUSEADDR, &opt, sizeof(opt));
4 addr.sin_port = htons(port); // 9999 or 10001
5 bind(fd, &addr, sizeof(addr));
6 listen(fd, 8);
7 // serve_loop: select() with 200ms timeout -> accept() -> detach thread
```

14 Continuous Simulation Launcher

The `scripts/run_sims.py` launcher enables scaled-up compute by spawning multiple `vsepr` viz workers with randomised initial conditions:

$$\text{worker}_k \leftarrow (\text{species}_k, T_{0,k}, T_{1,k}, N_k, \text{seed}_k), \quad k = 1, \dots, W \quad (31)$$

where W is the worker count (`--workers` flag, default 2).

Randomisation ranges:

- Species: Ar, N₂, CO₂, H₂, O₂, Ne, Xe, Kr, CH₄, H₂O
- Temperature: 5 intervals spanning [77, 1500] K
- Atom count $N \in \{32, 64, 125\}$ (simple cubic fill volumes)
- Seed: uniform random on $[1, 10^6]$

Failed or completed workers are automatically restarted after a configurable delay, providing indefinite compute coverage for Monte Carlo structural sampling across the full temperature–species phase space.

15 Extension to Thermal Pipe, Gas, and Crystalline Codes

The dual-port pattern applies uniformly to all simulation modules. Any module that produces an atom-level state and a set of scalar metrics can feed both ports by:

1. Constructing an `AtomicFrame` from the live atom array (fill `id`, `Z`, `r`, `v`, `q`, `state-tag`, `chi`)
2. Constructing an `AnalysisFrame` from accumulated history and property estimates
3. Calling `VizServer::push_atomic()` and `push_analysis()` at the appropriate cadence

For crystalline modules: set `lattice.active = true` and populate `a`, `b`, `c` from the unit cell. Bond type $\tau = 2$ (crystal adjacency) should be used for lattice neighbours. Defect atoms receive $s_i = 2$ so the Atomic View highlights them in the defect colour mode.

For thermal pipe modules: q_i carries the local heat flux or temperature deviation; χ_i encodes the thermal stress fraction. The Analysis View property panel naturally exposes k_{eff} , c_s , and α_T from the same `PropertySnapshot` struct.

Guiding principle:

From electron to dust, all properties are a must. Every atom, at every scale, carries the same record. Every property, at every phase, appears in the same panel.

16 Summary

Component	Description
<code>include/core/viz_server.hpp</code>	Complete API: AtomRecord, BondRecord, LatticeRecord, AtomicFrame, AnalysisFrame, PropertySnapshot, VizServer, generator declarations
<code>src/core/viz_server.cpp</code>	CPK colour table, JSON serialisers, gas2 frame generators (MB sampling, analytic $f(v)$, EOS candidates), TCP socket server, CLI dispatch
<code>tools/viz_atomic.py</code>	Port 9999 viewer: 3D projection, depth sort, 5 colour modes, bonds, lattice wireframe, HUD, mouse/keyboard controls
<code>tools/viz_analysis.py</code>	Port 10001 viewer: 2×3 panel grid (timeline, $f(v)$, 3D crystal, property bars, candidate comparison, KE histogram + crystal metrics)
<code>scripts/run_sims.py</code>	Continuous random sim launcher: randomised species/T/N/seed, auto-restart, W parallel workers